# Basil C K

*AI Engineer Intern | AI & Machine Learning | Computer Vision*

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Kochi, Kerala, India

## Profile

AI Engineer Intern with hands-on experience developing AI-powered applications using Python, FastAPI, Computer Vision, LLM integrations, and REST APIs. Passionate about building scalable AI systems from model development to deployment for real-world applications.

## EDUCATION

**B.Tech in Artificial Intelligence and Data Science**
*JCT College of Engineering and Technology, Anna University*
2022 – 2026
CGPA: 8.47 / 10

## TECHNICAL SKILLS

**Languages**
- Python, SQL

**AI & Machine Learning**
- Computer Vision, Deep Learning, CNNs, YOLO, OpenCV, TensorFlow/Keras, PyTorch, Model Training & Evaluation

**LLM & Generative AI**
- LLM Integration, Prompt Engineering, RAG Fundamentals, AI Chatbot Development, OpenAI-Compatible APIs, Conversational AI, AI Workflow Automation

**Backend Development**
- FastAPI, Flask, REST APIs, API Design, Backend Development, Async Programming

**Data & Databases**
- NumPy, Pandas, SQL

**Tools & Platforms**
- Git, Docker, Linux, VS Code, Jupyter Notebook

**Cloud & DevOps**
- OCI AI Foundations, AWS Fundamentals, CI/CD Fundamentals

## EXPERIENCE

**AI Engineer Intern** — 06/2026 – Present
*Difinity Digital* — Kochi, Kerala

- Developing AI-powered applications using Python, FastAPI and REST APIs.
- Building Computer Vision, Generative AI and workflow automation solutions for real-world clients.
- Designing backend services, AI integrations and production-ready APIs while collaborating with cross-functional teams.

## PROJECTS

**Hybrid CNN–YOLO Framework for Car Damage Severity Classification and Localization** 🔗
*Python | TensorFlow/Keras | YOLO | OpenCV | FastAPI | Computer Vision*
• Developed an end-to-end hybrid CNN–YOLO framework for vehicle damage detection, localization and severity classification.
• Built the complete pipeline including dataset preparation, model training, evaluation and a FastAPI inference backend.
- Presented the research at the **2026 IEEE International Conference on Natural Language Processing and Computer Vision (ICNPCV)**, demonstrating the proposed hybrid framework.

**Vexilon – AI-Powered Chrome Assistant** 🔗
*Python | Flask | REST APIs | LLM APIs | Chrome Extension | Git*
• Developed an AI-powered Chrome extension with a Flask backend for conversational assistance.
• Integrated LLM APIs using OpenAI-compatible interfaces and designed REST APIs.
• Implemented modular session-based interaction workflows.

**Vocalix – AI Voice Assistant** 🔗
*Python | Speech Recognition | LLM APIs | Automation Libraries*
• Built a voice-controlled AI assistant using speech recognition and LLM APIs.
• Automated system-level tasks through speech-to-command processing.
• Designed a modular real-time interaction pipeline.

## PUBLICATIONS

**A Hybrid CNN–YOLO Framework for Car Damage Severity Classification and Localization**
2026 *IEEE International Conference on Natural Language Processing and Computer Vision* (*ICNPCV*)
- Published and presented at the IEEE ICNPCV 2026 conference held at Global Academy of Technology, Bengaluru, India.
- Research focused on a hybrid deep learning framework integrating YOLO-based object detection with CNN-based damage severity classification for automated vehicle damage assessment.

## ADDITIONAL LEARNING & CERTIFICATIONS

• **IBM Data Science** – Selected Coursework (Python, Data Analysis)
• **Oracle Cloud Infrastructure 2023** – AI Foundations Associate
• **Google AI Essentials**

## LANGUAGES

English | Malayalam | Tamil